

DNS on the Host Cheat Sheet

How a Linux server turns a name like example.com into an IP address, and how to check and fix it.

How a name gets looked up

```
app (ping, curl, apt) -> /etc/nsswitch.conf ← "hosts: files dns" = hosts file first, then DNS
-> 1. files: /etc/hosts ← found here? done, DNS is never asked
-> 2. dns: /etc/resolv.conf says nameserver 127.0.0.53
-> systemd-resolved (local stub + cache) ← answers from its cache, or asks...
-> upstream DNS servers, e.g. 192.168.1.1 1.1.1.1 ← set per interface (netplan / nmcli / DHCP)
```

Using it

Command	Does	Command	Does
resolvectl status	DNS servers per link	getent hosts example.com	Looks up like apps do
resolvectl query nas	Looks up via resolved	dig example.com	DNS only: skips hosts file
resolvectl flush-caches	Forgets cached answers	dig @1.1.1.1 example.com	Asks one server directly
resolvectl statistics	Cache hits and misses	ls -l /etc/resolv.conf	Who manages it (below)

Reading resolvectl status

```
Global
  resolv.conf mode: stub ← /etc/resolv.conf points at 127.0.0.53 (normal)
Link 2 (eno1)
  Current Scopes: none ← bridge port: no IP, so no DNS (normal)
Link 3 (main)
  Current Scopes: DNS ← this interface is used for DNS
  Protocols: +DefaultRoute -LLMNR -mDNS -DNSoverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.1.1 ← the one in use right now
DNS Servers: 192.168.1.1 1.1.1.1 ← all servers for this link
DNS Domain: example.com ← search domain: "nas" tries nas.example.com
```

+DefaultRoute = this link answers every name not claimed by another link's domain.

/etc/resolv.conf: who owns it?

Is -l shows	Means
-> stub-resolv.conf	Normal (Ubuntu): ask 127.0.0.53
-> resolve/resolv.conf	resolved, real servers listed
A plain file	NetworkManager (RHEL) or by hand

/etc/hosts (checked before DNS)

```
127.0.0.1    localhost
127.0.1.1    web1.example.com web1
192.168.1.20 nas.example.com nas
203.0.113.10 test.example.com
```

IP, full name, then short aliases. Works at once.

Setting DNS permanently (editing /etc/resolv.conf won't last)

```
nameservers: {addresses: [192.168.1.1, 1.1.1.1], search: [example.com]} ← Netplan, under main
nmcli con mod main ipv4.dns "192.168.1.1 1.1.1.1" ipv4.ignore-auto-dns yes ← then con up
DNS=1.1.1.1      in /etc/systemd/resolved.conf ← all links; restart systemd-resolved
```

Split DNS: different servers for different names

```
resolvectl dns wg0 203.0.113.53 ← the VPN link wg0 gets its own DNS server
resolvectl domain wg0 '~corp.example.com' ← ~ = send only *.corp.example.com there · temporary
```

Common problems

Symptom	Likely cause and fix
"Temporary failure in name resolution"	No usable DNS server: resolvectl status shows no Current Scopes: DNS
Apps and dig give different answers	An /etc/hosts entry wins for apps: compare with getent hosts
Still getting the old IP	resolvectl flush-caches (upstream may cache until the TTL ends)
Short names like nas fail	No search domain: add search: [example.com]

Remember: test with **getent hosts**: it follows the same path as your apps, **/etc/hosts** included. **dig** doesn't.